

1. Build Drug SE Similarity matrix

	D1	D2	Dn
D1	1	0.5	-
D2	-	1	-
D3	0.3	0.6	1
Dn	-	0.7	-

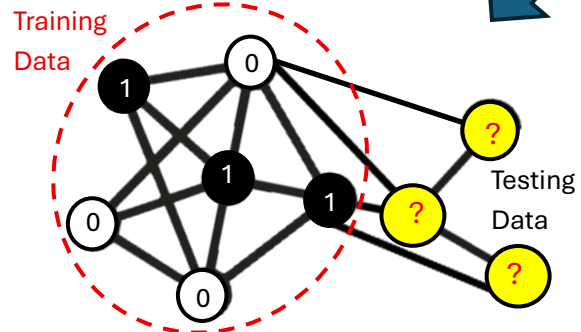
Drug SE Similarity
Network (NB)

Drug gene Similarity
Network (NA)

2. Build Drug Gene Similarity matrix

	D1	D2	Dn
D1	1	0.7	-
D2	-	0.3	0.89
D3	0.1	-	0
Dn	-	0.3	-

3. For a given SE (E.g. Pruritis)



1 = SE present, 0 = SE absent,
? = Predict SE based on training data

How NB depends on NA??

4. Rank SE by predictability

Classifier

Ranked SE	Score
Side Effect 1	0.95
Side Effect 2	0.87
Side Effect 3	0.67
.....	
Side Effect n	0.00